# History of Science & Technology in Modern India

# Introduction

•**Overview:** Transition of science in India from a tool of colonial interest to a vehicle for national identity.

•**Key Focus:** * Involvement of Indians as assistants in British scientific machinery.

•The "Indian-Response Stage," where natives took initiative in research.

•**Learning Objective:** To trace the history of scientific association in colonial India and its role in nation-building.

# Involvement of Indian in Early Scientific development:
## The "Moonshee" Phase

**"Moonshee Phase"** (approx. 1770s–1830), a specific era in colonial Indian history characterized by the British East India Company's heavy reliance on local intellectuals to map and understand a subcontinent they had conquered but did not yet know.

- **Strategic Necessity:** In the late 18th century, the British lacked the geographical and cultural knowledge required for military and administrative control. They employed "natives"—specifically **Munshis** (educated Muslims) and **Pandits** (educated Hindus)—to survey roads, gather intelligence on rivals like the Marathas, and correct erroneous European maps.

**1. The Paradox of Necessity vs. Fear**

- The British required more than just military power; they needed **"The Lay of the Land."**
- **The Utility:** Native officers, Munshis, and Pandits were the primary data collectors. They mapped the Deccan, explored the roads to Kabul and Multan, and corrected European geographical errors (such as the actual path of the Ghaggar river).
- **The Anxiety:** The Company feared that "Properly Instructed Indians" would become too knowledgeable and potentially sell information to French or Dutch rivals. By 1813, the British officially banned "harkaras" (messengers/assistants) from survey work to prevent Indians from acquiring technical expertise.

**2. Significant Native Contributors**

- While the British names (Reynolds, Wilford, Burrow) are well-known, the text highlights the essential work of the Indians behind the maps:
- **Golam Mohamad (1774):** Explored the Deccan and gathered intelligence on the Maratha powers.
- **Mirza Mogul Beg (1786–1796):** Collected the vast data used to map the countries west of Delhi.
- **The "Anonymous" Experts:** Col. Charles Reynolds employed a team for 12 years to map Western India, valuing their work so highly he offered to pension them off himself.

### 3. Education as an Information Pipeline

- The establishment of the **Calcutta Madrasa (1781)** and the **Banaras Sanskrit College (1791)** was not an altruistic attempt to modernize Indian education. Instead, it was strategic:
- **Language Barrier:** English was intentionally **not** taught. The British wanted "non-English knowing Indians" to remain in their traditional spheres.
- **Knowledge Extraction:** These institutions were designed to produce a "band of young Hindus and Muslims" who could extract traditional laws, customs, and geographical knowledge from their elders and translate it for their new rulers.

### 4. The Erasure of the "Moonshee"

- By 1830, the "Moonshee Phase" ended. The British moved toward a system where:
- Only **covenanted or military officers** could handle map-making.
- Manpower shortages were solved by training **European orphaned boys** (via the Madras Surveying School) rather than Indians.

# The "Baboo" Phase

The **"Baboo Phase"** marks a pivotal shift in colonial policy (post-1817), where the British moved from keeping Indians in a state of ignorance to providing them with a "calculated" English and scientific education. This was not born of altruism, but of administrative and economic necessity.

Here is a summary of the key themes from this phase:

**1. From Ignorance to Instruction**

- Following the final defeat of the Marathas in 1818, British supremacy was unassailable. Governor-General Lord Hastings and later Lord William Bentinck realized that the Empire had grown too large to be managed by Europeans alone. The "Baboo Phase" aimed to create a class of Indians who were Westernized in outlook and could serve as the "cheapest agency" for administrative and scientific tasks.

**2. The Rise of the "Computers"**

- The Great Trigonometrical Survey (GTS), led by **George Everest**, faced a massive backlog of data. While the British handled the prestigious field work, they recruited brilliant students from the newly established **Hindu College (1817)** to serve as "Computers" (calculators).
- **Radhanath Sickdhar:** The most famous of these recruits, Sickdhar was an exceptionally brilliant mathematician who rose to Chief Computer. While a legend suggests he calculated the height of Mt. Everest, the text clarifies his role was peripheral to the final calculation, though he was "thoroughly European in outlook."
- **Economic Exploitation:** Most native computers remained underpaid (stagnating at ₹40 p.m.), leading many to quit for better-paying administrative roles like Deputy Collectors.

**3. The "Clandestine" Surveyors**

- While Indians were generally barred from high-level field surveys, the British utilized them for **high-risk, surreptitious missions** in trans-Himalayan regions (Tibet/Central Asia).
- **Strategic Utility:** Native explorers could blend into hostile territories where a European would be killed.
- **Intentional Limitation:** These explorers (often referred to only by initials or titles like **"The Mullah"** or the **"Pandit brothers"** Nain and Kishan Singh) were taught to take observations but **not** to reduce the data. This ensured they remained dependent tools of the British and could not "cheat" or use the information independently.

## 4. Educational Glass Ceilings

- The science taught at institutions like Hindu College (later Presidency College) was intentionally shallow. **Mahendralal Sircar** noted in 1855 that the primary goal was literacy in English rather than true scientific mastery. Similarly, observatories established by Indian royalty (Lucknow, Hyderabad) remained under European control, preventing any real transfer of astronomical knowledge to Indians.

## 5. Evolution of the Native Role

- By the end of this phase, the Indian's role in the British scientific machinery had evolved through three tiers:
- **Coolies:** Manual labor (Moonshee Phase).
- **Calculators:** Data processing and "computing" (Early Baboo Phase).
- **Professional Assistants:** Gradually graduating to become doctors and engineers to maintain the Empire’s new infrastructure (railways, telegraphs, and canals).

**6. The "Cheaper and Efficient" Policy**

- Governor-General **Lord William Bentinck's** 1829 minute was a masterclass in colonial pragmatism. He recognized that the Empire's administrative "business" was becoming too expensive to be handled solely by Europeans. By calling for an "enlarged employment of native agency," he wasn't advocating for Indian empowerment, but for **fiscal optimization**.

**7. George Everest and the Data Crisis**

- By 1830, when **George Everest** took the helm of the Survey, the department was in crisis. There was an **eight-year backlog** (arrears) of raw field data. The British officers were excellent at the "heroic" work of physical surveying (fieldwork), but they were overwhelmed by the mathematical "drudgery" of data reduction.

**8. The Birth of the "Human Computer"**

- To solve this, a structural wall was built:
- **The Field Staff:** Remained the domain of the British (prestige and exploration).
- **The Computing Office:** Created specifically for Indians (mathematics and data processing). This was the first time "Computing" was established as a distinct professional role for Indians under the British.

**9. The Role of Hindu College**

- The text highlights **Hindu College** as the "factory" for this new class of professionals. It was ready to fulfill the Government's hope because it was already producing English-literate, mathematically trained students who were willing to work for the low wages the Company offered.

1. **The Practicality of Involvement**

- Because India was too densely populated for "white settlements" (unlike Australia or North America), the British realized they couldn't rule by sheer numbers alone. They had to co-opt the local population into the machinery of government.

**2. The Shift from "Ruler" to "Law"**

- The establishment of the **Supreme Court of Justice at Calcutta (1774)** marked a pivot in Indian history:
- **Old System:** Justice often depended on the whim or personality of the monarch.
- **New System:** A framework of law existed independently of the individual in power.
- **Legacy:** This created the "Indian Lawyer" class, which produced many of the leaders of the later independence movement.

**The Secularization of Scholarship**

- The need to understand Hindu and Muslim law for judicial purposes led to a strange linguistic journey:
- Sanskrit →Persian →English.
- This process proved Sanskrit still had "utility value" and wasn't just a liturgical relic.
- This scholarship culminated in the founding of the **Asiatic Society (1784)** by Sir William Jones, marking the birth of modern Indology.
- **Key Themes in Early Scientific Exchange**
- **The "Black Box" of Indian Astronomy:** Figures like Cassini were stunned by the "Hindu Mathematical Tables." It suggests that Indian astronomers had developed highly sophisticated algorithms for predicting celestial events that worked perfectly in practice, even if they didn't align with European Newtonian physics at the time.
- **Administrative Synchronization:** The Madras Government’s funding of John Warren’s *Kala Sankalita* wasn't just a gesture of intellectual curiosity. It was a **logistical necessity**. To collect taxes (Revenue) and settle disputes (Judicial), the British needed to know exactly what day it was on a local calendar versus their own.
- **Cultural Validation:** The fact that the Pandits offered to pay for Warren’s daughter’s wedding is a profound detail. It suggests a rare moment of mutual respect where the British officer was seen not just as a ruler, but as a scholar who had "earned" his place in their intellectual circle.

## Initiation of Indian Response

- One of the most striking points in the text is the idea that Hindus regained self-esteem because their past was **"certified by the Europeans themselves."**
- **The Irony:** It took Western scholars like Sir William Jones or John Warren praising Indian mathematics and Sanskrit to make some Indians realize the scale of their own heritage.
- **The Result:** This "external validation" acted as a shield against the British narrative of "civilizing" a "backward" people. If India was once a world leader in science and philosophy, then its current subjugation was a temporary lapse, not a natural state.
- **The British "Four-Fold" Introduction**
- The text credits the first century of British rule with introducing four transformative elements:
- **English Language/Literature:** Provided a *lingua franca* for Indians from different regions to communicate and organize.
- **Western Thought:** Introduced concepts of "Liberty," "Equality," and "Democracy" (which Indians quickly pointed out the British weren't practicing in India).
- **India’s Glorious Past:** Re-discovered through Indology and archaeology.
- **Modern Science and Education:** Provided the technical tools for modernization.

**The Goal: The "World's Science Club"**

- The final sentiment is about **parity**. The goal for the Indian intelligentsia shifted from merely surviving under the Empire to proving they could compete on the global stage.
- "It was now for the Indians to prove... that they... were capable of becoming full-fledged members of the world‘s science club."
- This mindset paved the way for the likes of **Jagadish Chandra Bose**, **S.N. Bose**, and **C.V. Raman**—scientists who sought to excel in modern Western science while drawing confidence from their Indian roots.

**Ramachandra and Indian Mathematics**

- **The Significance of Ramchandra’s Career**
- **The Vernacular Advantage:** Unlike the "Anglicists" who believed science could only be taught in English, Ramchandra’s success at **Delhi College** proved that complex concepts like Calculus could be localized. This was a "revolutionary" pedagogical shift for 1844.
- **The Global Recognition:** His work, *Problems of Maxima and Minima* (1850), initially faced skepticism in India (the *Calcutta Review*). However, it found a champion in **Augustus De Morgan**, one of the giants of 19th-century mathematics.
  - De Morgan didn't just see a math book; he saw the **"restoration of the native mind."** He believed Indians had a natural aptitude for mathematics that had been dormant and could be reawakened.

## Key Milestones in Ramchandra's Life

| Year | Event | Significance |
|---|---|---|
| **1844** | **Teacher of European Science** | **Taught modern science in the vernacular at Delhi College.** |
| **1850** | ***Problems of Maxima and Minima*** | **His seminal work, later reprinted in England (1859).** |
| **1858** | **Post-Mutiny Appointment** | **Appointed to Roorkee Engineering College, showing the resilience of his career despite the 1857 Uprising.** |
| **1861** | ***A New Method of the Differential Calculus*** | **His second major contribution to mathematical theory.** |
| **1863** | **Tutor to Maharaja of Patiala** | **Transitioned from colonial service to influencing the education of Princely States.** |

## Industrial Art Society and R.L.Mitra

- The mid-19th century marked a critical shift from Indians being passive recipients of British education to becoming **vocal critics** of its limitations. This passage highlights a growing realization: the British were training Indians to be **cogs in the colonial machine** (clerks and accountants) rather than the architects of their own nation's industry.
- **The Critique of "Clerk-Making" Education**
- The *Hindu Patriot's* 1854 editorial is a stinging indictment of the colonial syllabus. It identifies two major flaws:
- **Lack of Agency:** The education was designed for "accountants or letter writers"—roles that serve an existing power structure but do not create new value.
- **Economic Stagnation:** The "children of the soil" were being denied the technical knowledge required to develop India's own natural resources, ensuring the country remained economically dependent on Britain.

**Science as a Social Disruptor**

- **Rajendralal Mitra** (1829–91) offered a brilliant sociological insight. He saw science and "industrial art" (technology/craft) as a way to break the **Caste System's** grip on Indian progress.
- **The Problem:** Traditionally, intellectual pursuit was the domain of the "higher" classes, while manual/industrial work was left to the "least educated" (the laboring castes). This created a wall between **theory** and **practice**.
- **The Solution:** By teaching "Industrial Art" to the educated classes, Mitra hoped to bridge this gap. If a scholar learned to be a technician, the social stigma of manual labor would vanish, and India could modernize.

**The "Parallel Infrastructure" Movement**

- Mitra's **Industrial Art Society** was a precursor to what would eventually become the **Swadeshi movement** in science. However, as the text notes, these early attempts faced significant hurdles:

| Challenge | Why it Failed Initially |
| --- | --- |
| Capital | Lack of independent funding outside the British government. |
| Scale | These were "few and half-hearted" compared to the massive British state machinery. |
| Apathy | A large section of the Indian elite was still content with "safe" government jobs. |

**The Indian Association for the Cultivation of Science**

- This is the moment the "Indian response" matured into a permanent institution. The story of **Dr. Mahendralal Sircar** and the founding of the **IACS** represents a sophisticated middle ground between total dependence on the British and a desire for intellectual sovereignty.
- Sircar's vision was notably different from Rajendralal Mitra's earlier focus on "industrial art." He wasn't interested in making a "nation of artisans"; he wanted a **nation of thinkers.**

**The Strategy of "Science Speculation"**

- The Bengali intelligentsia of the late 19th century possessed a unique confidence. Having mastered English literature (Shakespeare), they felt intellectually equal to their colonizers. Their strategy for science was two-fold:
- **The "Applied" burden:** Let the British government handle the expensive, utilitarian side of science (telegraphs, railways, canals).
- **The "Pure" pursuit:** Let Indians cultivate "Science Speculation"—the theoretical research that defines a civilization's intellectual rank.

**Dr. Mahendralal Sircar: The Reluctant Radical**

- Sircar's background as a "poor orphan" who rose through Western education made him a pragmatic leader.
- **Professional Defiance:** His MD was in Western medicine, but his shift to **Homeopathy** showed he was willing to endure "professional ostracism" for his beliefs—a trait that served him well when challenging the scientific status quo.
- **The Funding Masterstroke:** He understood a harsh reality of colonial India: the wealthy "Zamindars" and merchants wouldn't donate to a national cause unless it had a "stamp of approval" from the British. By securing the support of **Sir Richard Temple**, he effectively "unloosed the purse-strings" of the Indian elite.
- **The Birth of IACS (January 1876)**
- The **Indian Association for the Cultivation of Science** was revolutionary because it was the first institute of its kind **funded by Indians, for Indians.**

| **Feature** | **IACS (Sircar's Vision)** | **British Government Colleges** |
|---|---|---|
| **Goal** | Original discovery and research. | Producing clerks and subordinates. |
| **Funding** | Private Indian donations. | Government Treasury. |
| **Philosophy** | Mastery of Western principles to eventually surpass them. | Practical utility for the Empire. |

**1. The Industrial Spark: Tata Steel (1911)**

- The founding of Jamshedpur is a perfect example of the **"Science-Industry Alliance"** that the earlier reformers had dreamed of.
- **The Scientist: P.N. Bose**, a geologist who, after retiring from the British-led Geological Survey of India, used his expertise for an Indian Princely State (Mayurbhanj).
- **The Industrialist: Jamsetji Tata**, who had the vision and capital to act on Bose’s data.
- **The Result:** The establishment of the Tata Steel Mill in 1911 was a massive blow to the "nation of accountants" narrative. It proved Indians could manage heavy industry—the very backbone of a modern nation—independent of British manufacturing.

**2. The Astrophysical Correction: Naegamvala vs. Lockyer**

- The story of **Kavasji Dadabhai Naegamvala** is a classic "student surpasses the master" tale. It highlights the growing intellectual independence of Indian scientists:
- **The Conflict:** Even though Naegamvala was a protégé of the legendary **Sir Norman Lockyer** (who discovered Helium), he did not blindly follow his mentor’s theories.
- **The "Scientific Rebellion":** Naegamvala used the Poona Observatory to disprove Lockyer’s hypothesis regarding nebular lines in Orion. This was a "full-fledged member of the science club" (as your previous text phrased it) doing exactly what science demands: challenging established authority with better data.
- **Funding Hybridity:** The fact that half the money for the Poona observatory came from **Indian donations** underscores Mahendralal Sircar's earlier strategy—using private Indian wealth to build scientific infrastructure.

- The founding of the **Indian Institute of Science (IISc)** in Bangalore (1911) represents the "crowning achievement" of the era you’ve shared. It was the moment Indian private capital, princely support, and scientific ambition converged to create an institution that bypassed the limitations of the British colonial university system.

**3. The Nomenclature: "Institute" vs. "University"**

- Jamsetji Tata’s choice of the name **"Indian Institute of Science"** was a deliberate critique of the existing British education system.
- **The Problem:** At the time, "Universities" in India (like Calcutta, Bombay, and Madras) were primarily **examining bodies**. they didn't conduct research; they just certified clerks and lawyers.
- **The Vision:** Tata wanted a center for **original research and postgraduate study**. By calling it an "Institute," he signaled a departure from rote learning toward the "Science Speculation" Mahendralal Sircar had championed.

**4. The Parsi-Mysore Alliance**

- The establishment of IISc was a masterclass in **inter-regional Indian cooperation**:
- **The Capital:** "Parsi money" from the Tata family in Bombay was invested far outside their own community’s geographic base.
- **The Land and Support:** The **Maharaja of Mysore** provided 371 acres of land and significant financial grants.
- **The Facilitator: Dr. Hormusji Bhabha** (father of the famous physicist Homi J. Bhabha) served as the vital link between the Mysore state and the Tata family.

**5. The Colonial Paradox: Indian Money, British Control**

- Even though the initiative and funding were Indian, the text points out a persistent reality of 1911: **The control remained British.**
- The first Director, **Morris Travers**, and much of the early faculty were British.
- This created a "hybrid" space where Indian students were trained to the highest global standards, but still operated under the administrative umbrella of the Raj.

## University College of Science and Technology: Sir Asutosh Mookerjee

### 1. Sir Asutosh Mookerjee: The "Tiger of Bengal"

- Mookerjee’s life trajectory mirrors the very "Indian Lawyer" class mentioned in your first excerpt.
- **The Mathematician-turned-Jurist:** He was a brilliant mathematician first, but the lack of opportunities for Indians in colonial academia forced him into Law—a common theme where India’s best scientific minds were diverted into the legal framework the British had built.
- **The Visionary Vice-Chancellor:** He used his prestige as a High Court Judge to execute a "hostile takeover" of the academic landscape, transforming Calcutta University from a mere examining body into a world-class research hub.

## 2. The Great Sacrifice: C.V. Raman

- The story of C.V. Raman's transition from a government accountant to a professor is one of the most famous legends in Indian science:
- **The Part-Timer:** Working at the **IACS** (Mahendralal Sircar's legacy) before and after his office hours, Raman proved that the "Science Speculation" dream was viable even on a shoestring budget.
- **The Choice:** When Mookerjee offered him a professorship, Raman took a **nearly 50% pay cut** (from 1100 to 600 rupees). This highlights a shift in the Indian psyche: the pursuit of scientific truth and national prestige had become more valuable than a secure, high-paying colonial "clerkship."

## 3. Financial and Racial Sovereignty

- Mookerjee's strategy for the University College was a masterstroke of institutional design:
- **Lawyers Funding Science:** Just as Sircar used British approval to get money, Mookerjee used the wealth of the "Indian Lawyer" class (like Taraknath Palit and Rashbehari Ghosh) to fund the college.
- **The Racial Clause:** Crucially, he ensured the professorships were reserved for **Indians themselves**. This was a direct challenge to the "British-controlled" model of the IISc and other government colleges.

- **Path breaking scientific innovation: Raman, Shah and Bose**

The Statistical "Takeoff" (1920–1929)

The data from the 1920s reveals a highly sophisticated, internationally-connected Indian scientific community:

**Educational Sovereignty:** More than half ($53\%$) of the physics doctorates were earned within **Indian Universities**. This proves that the infrastructure built by Sir Asutosh Mookerjee and others was now self-sustaining.

**Global Validation:** The fact that the majority of citations came from the **US and Germany** (rather than just the UK) is critical. It shows that Indian science was being judged—and praised—by the best minds in the world, independent of the British colonial lens.

- **Srinivasa Ramanujan**
- For the aspiring Indian scientists of the 1920s and 30s, Ramanujan was the ultimate proof of **innate superiority**.
- If a "college dropout" from Kumbhakonam could stun the greatest mathematicians at Cambridge, then the British argument of "intellectual tutelage" was dead.
- He became a symbol: if an Indian mind could conquer the most abstract of all sciences (Mathematics) without formal training, then India was ready for self-rule in every other domain.

## Science in Post Independent Era

- **The Colonial Trauma and the "Application" Gap**
- A profound historical insight is offered regarding why Indians favored **theory** over **application**:
- **Science as a Destroyer:** Indians didn't see science as a tool for wealth; they saw it as the weapon the British used to destroy Indian artisans and textiles.
- **Class Divide:** The new English-educated middle class (lawyers, scholars) was disconnected from the traditional manufacturing/artisan classes.
- **The Result:** The elite celebrated "Science Speculation" (which felt noble and intellectual) but ignored "Science Application" (which felt like the manual labor of the "lower" classes or the machinery of the oppressor).